SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-020: Student Password Change from Student Dashboard

Field	Details
Change Request ID	CR-020
Change Request Title	Student Password Change from Student Dashboard
Student Name	K.Varshika
Student ID	JF/ICT/24/10
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Students needed a secure and convenient way to change their own account passwords from the Student Dashboard. The change was required to improve account security and allow students to update their passwords without administrator assistance.

The system must verify the student's current password before accepting a new password. It must also validate the new password, confirm that both new-password fields match, and store the updated password securely as a hash in the database.

1.2 Existing Behaviour

Before implementation:

- Students could log in using their registered credentials.
- No dedicated password-change facility was available in the Student Dashboard.
- Students could not conveniently update their own passwords through the dashboard.
- Password authentication depended on the existing user authentication system.

1.3 Requested Change

A password-change facility was required in the Student Dashboard.

The functionality must:

- Allow only authenticated students to change their own passwords.
- Require the current password.
- Require a new password and confirmation password.

- Verify that the current password is correct.
- Check that the new password and confirmation password match.
- Require a minimum of eight characters, including uppercase letters, lowercase letters, numbers, and special characters.
- Reject a new password that is the same as the current password.
- Store the new password as a bcrypt hash.
- Display appropriate success and error notifications.
- Provide a Clear button to empty the password fields and return focus to the password input.
- Allow the student to log in using the new password after a successful change.
- Reject the old password after a successful password update.
- Preserve existing administrator functions, student CRUD operations, and authentication functionality.

2. Implementation

2.1 Implementation Summary

CR-020 was implemented to provide a secure password-change facility for students through the Student Dashboard.

The frontend collects the current password, new password, and confirmation password. The backend validates the submitted information, verifies the current password using bcrypt, checks the new password requirements, and updates the password hash in the users table.

The password-change request is protected by authentication middleware. The backend identifies the authenticated student's account and updates that account's password rather than allowing the student to select another user's account.

The interface also provides success and error feedback and a Clear button for resetting the password form.

2.2 Affected Files

File	Change	Purpose
`frontend/src/components/ChangeStudentPassword.jsx`	Added / Modified	Provides the password-change form, validation feedback, and Clear functionality.
`frontend/src/pages/StudentDashboardPage.jsx`	Modified	Integrates the password-change component into the Student Dashboard.
`frontend/src/styles/changePassword.css`	Added / Modified	Styles the password form, buttons, and

		notification elements.
`frontend/src/services/api.js`	Verified / Modified if required	Sends authenticated API requests and handles API errors.
`backend/src/routes/studentRoutes.js`	Modified	Registers the authenticated password-change endpoint.
`backend/src/controllers/studentController.js`	Modified	Implements password validation, current-password verification, and the password-change process.
`backend/src/models/studentModel.js`	Modified	Updates the user's password hash in the database.
`backend/src/middleware/authMiddleware.js`	Verified	Ensures that protected requests are authenticated.

Note: Retain only the files that were actually added or modified in the final report.

2.3 Technical Changes

A. Frontend Implementation

A password-change form was added to the Student Dashboard.

The form contains the following fields:

- Current Password
- New Password
- Confirm New Password

The frontend sends the entered values to the backend through the password-change API endpoint. It also displays success or error messages according to the server response.

The Clear button resets the form fields and returns focus to the password input.

B. Backend Validation

The backend validates the password-change request before updating the database.

The validation checks include:

- All required password fields are provided.
- The new password and confirmation password match.
- The current password matches the stored password hash.
- The new password satisfies the password-strength requirements.
- The new password is different from the current password.
- The request belongs to an authenticated student.

Invalid requests are rejected with an appropriate error response.

C. Password Security

The new password is hashed using `bcryptjs` before being saved.

The database stores the password hash rather than the plain-text password. The current password is verified by comparing the submitted password with the stored hash.

The password-change operation updates the `password_hash` field in the existing `users` table.

D. API Endpoint

The password-change functionality uses the following endpoint:

```
PATCH /api/students/me/password
```

The request requires a valid authentication token and accepts the current password, new password, and confirmation password.

E. Database Changes

The existing `users` table is used to store the updated password hash.

No separate password table is required for this feature. The existing user ID and authentication structure are retained.

F. Existing Functionality Preservation

The change is limited to the student password-change feature.

Existing login functionality, administrator authorization, student creation, student editing, student deletion, student profile functionality, and other existing CRUD operations remain unchanged.

2.4 Implementation Logic

1. The student logs in and opens the Student Dashboard.
2. The student opens the password-change form.

3. The student enters the current password.
4. The student enters the new password and confirmation password.
5. The frontend submits the form to `PATCH /api/students/me/password`.
6. The backend verifies the authentication token.
7. The backend validates the required fields and confirms that the new password fields match.
8. The backend verifies the current password against the stored bcrypt hash.
9. The backend checks the new password's strength and confirms that it differs from the current password.
10. The backend hashes the new password using bcrypt.
11. The model updates the password hash in the database.
12. The frontend displays a success notification.
13. The student can log in with the new password, while the old password is rejected.

2.5 Error and Data Handling

The implementation handles invalid password-change requests using appropriate validation and error responses.

The following conditions are handled:

- Missing required password fields.
- Incorrect current password.
- New password and confirmation password mismatch.
- Password that does not meet the strength requirements.
- New password identical to the current password.
- Missing or invalid authentication token.
- Database update failure.

The password form can be cleared so the student can enter the information again. Password values are not returned in the API success response.

3. Testing and Verification

3.1 Testing Approach

The implementation is intended to be verified through manual frontend testing and backend API testing.

Testing types:

- Functional testing
- Positive testing
- Negative testing
- Password validation testing
- Authentication and authorization testing

- Database update verification
- UI testing
- Error-handling testing
- Regression testing
- Login verification

Environment: Development
Browser: Google Chrome
Test Role: Authenticated Student
Backend: Node.js and Express.js
Database: MySQL
Frontend: React and Vite

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Change password using valid current and new passwords	Password hash is updated successfully	Password changed successfully	Pass
TC-02	Enter an incorrect current password	Password update is rejected	Appropriate error message displayed	Pass
TC-03	Enter different values in the new-password and confirmation fields	Password update is rejected	Password mismatch message displayed	Pass
TC-04	Enter a password shorter than eight characters	Password update is rejected	Validation message displayed	Pass
TC-05	Enter a password without the required uppercase, lowercase, number, or special character	Password update is rejected	Password-strength validation displayed	Pass
TC-06	Enter the same password as the current password	Password update is rejected	Appropriate validation message displayed	Pass
TC-07	Leave a required password field empty	Password update is rejected	Required-field validation displayed	Pass
TC-08	Click the Clear button	All password fields are emptied and focus returns to the password input	Form cleared successfully	Pass
TC-09	Submit the request without a valid authentication token	Access is rejected	Authentication error returned	Pass
TC-10	Log in using the new password after a successful change	Login succeeds	Login succeeds with the new password	Pass
TC-11	Attempt to log in using the old password after a successful change	Login is rejected	Old password rejected	Pass
TC-12	Verify that the database stores a password hash rather than plain text	The new password is stored as a bcrypt hash	Password hash updated in the database	Pass

TC-13	Verify that a student cannot change another student's password through this endpoint	Only the authenticated student's own password can be changed	Own-account restriction verified	Pass
TC-14	Verify existing administrator student CRUD operations	Existing add, edit, and delete functions continue to work	CRUD functions work correctly	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	14
Passed	14
Failed	0
Pass Rate	100%

Overall Test Result: Pass

The test counts and actual results above should be retained only after all 14 test cases have been executed successfully.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Authenticated students can access the password-change functionality.	Student dashboard and authentication test	Met
AC-002	The current password must be correct before the password is changed.	Incorrect-current-password test	Met
AC-003	The new password and confirmation password must match.	Password mismatch test	Met
AC-004	The new password must satisfy the required strength rules.	Password validation tests	Met
AC-005	A new password identical to the current password is rejected.	Same-password test	Met
AC-006	The password is stored securely as a bcrypt hash.	Database and code verification	Met
AC-007	The Clear button empties the form and restores input focus.	UI testing	Met
AC-008	Appropriate success and error notifications are displayed.	Positive and negative testing	Met
AC-009	The new password works for login and the old password is rejected.	Login verification	Met
AC-010	A student cannot use this endpoint to change another user's password.	Authentication and account-ownership verification	Met
AC-011	Existing administrator CRUD and authentication functionality remain operational.	Regression testing	Met

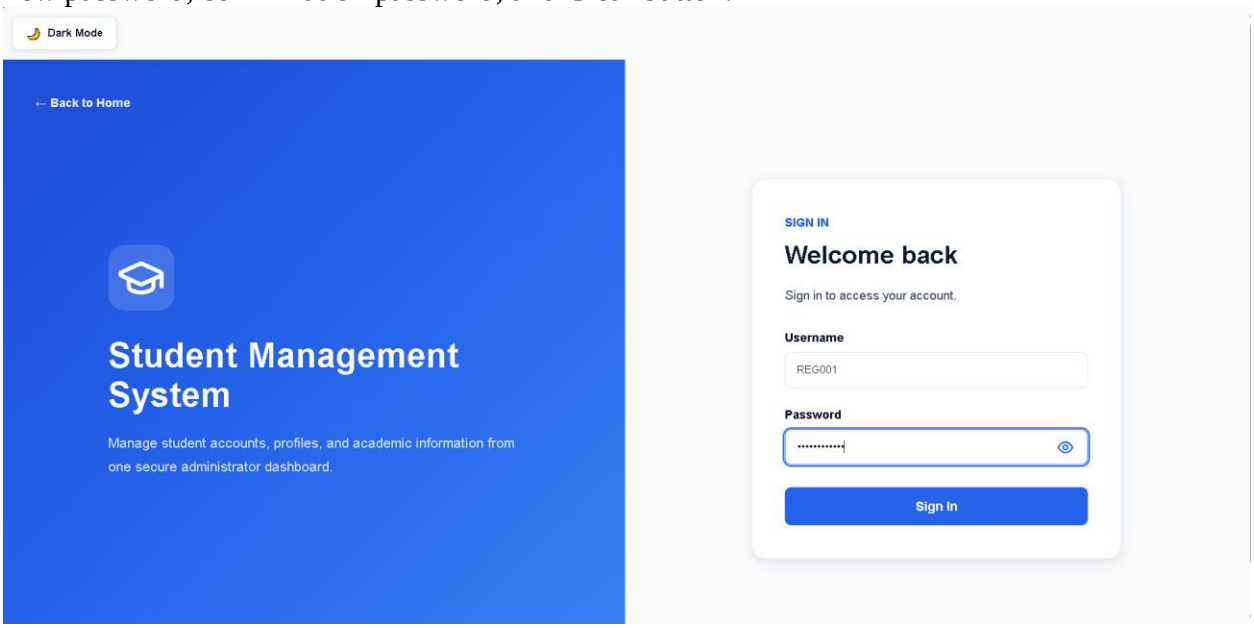
5. Evidence

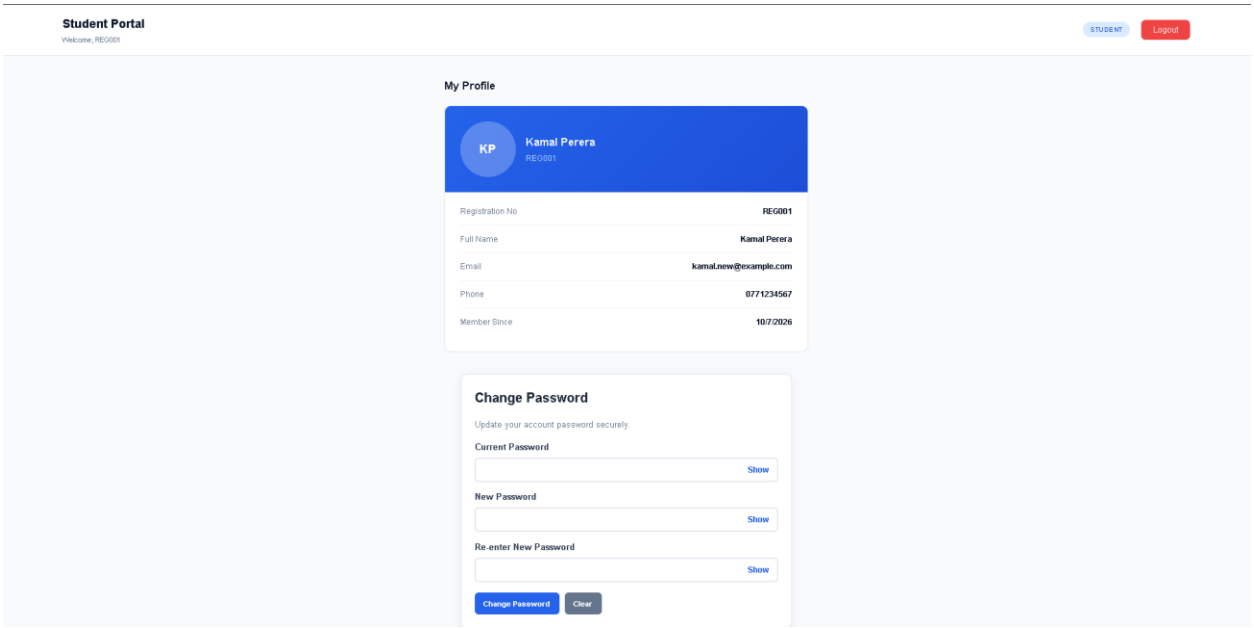
A maximum of two relevant screenshots should be attached.

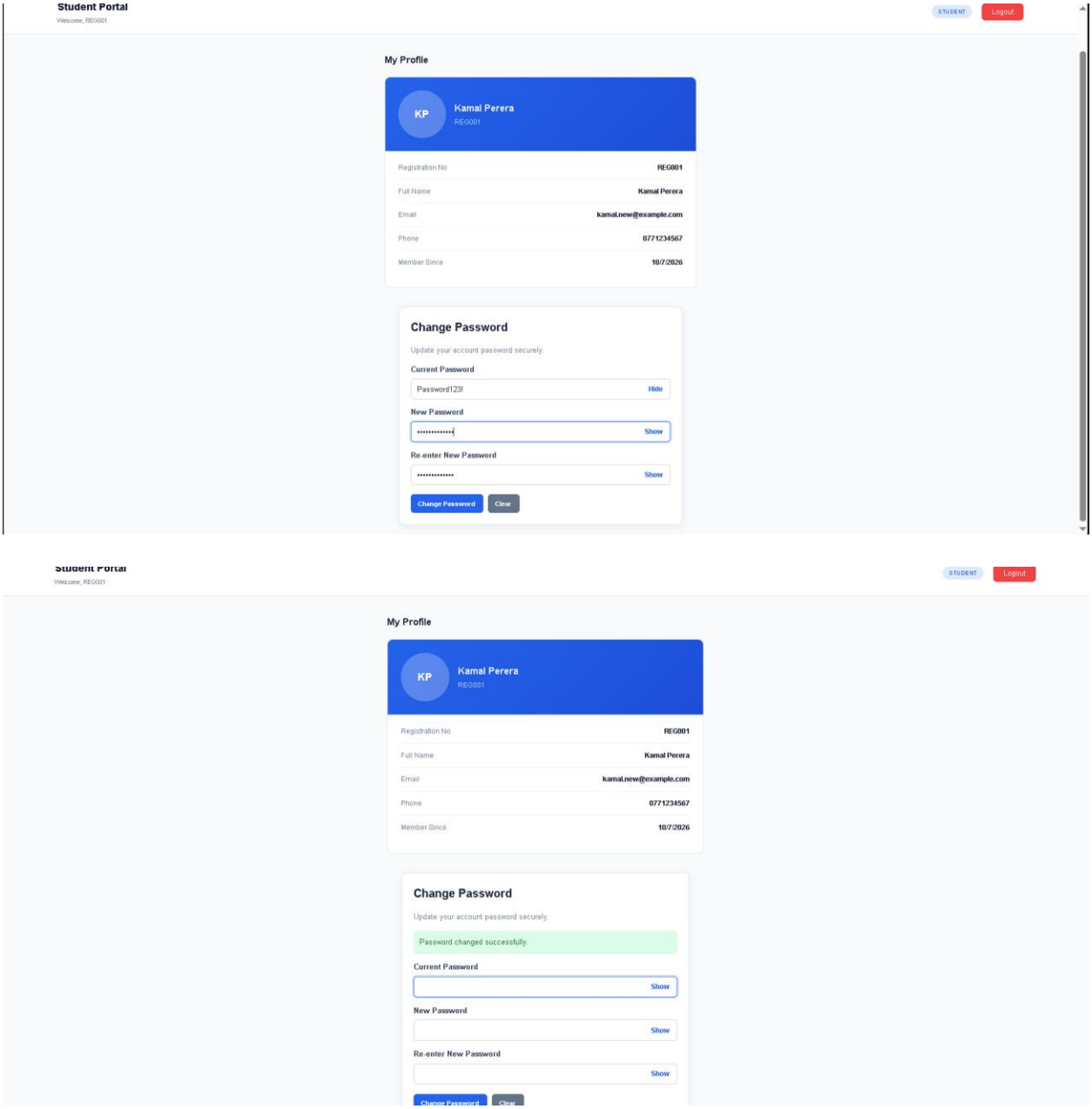
Evidence 01 – Student Password-Change Interface

Related criteria: AC-001, AC-003, AC-004, AC-007, AC-008

Figure 1: Student Dashboard displaying the password-change form with the current password, new password, confirmation password, and Clear button.



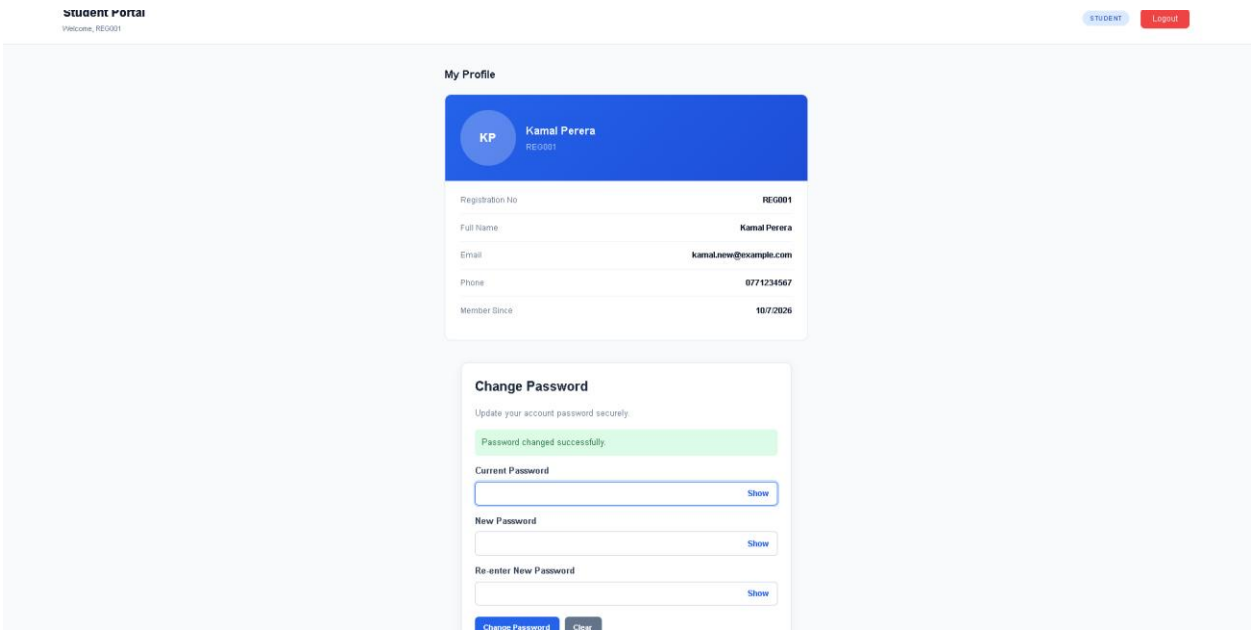


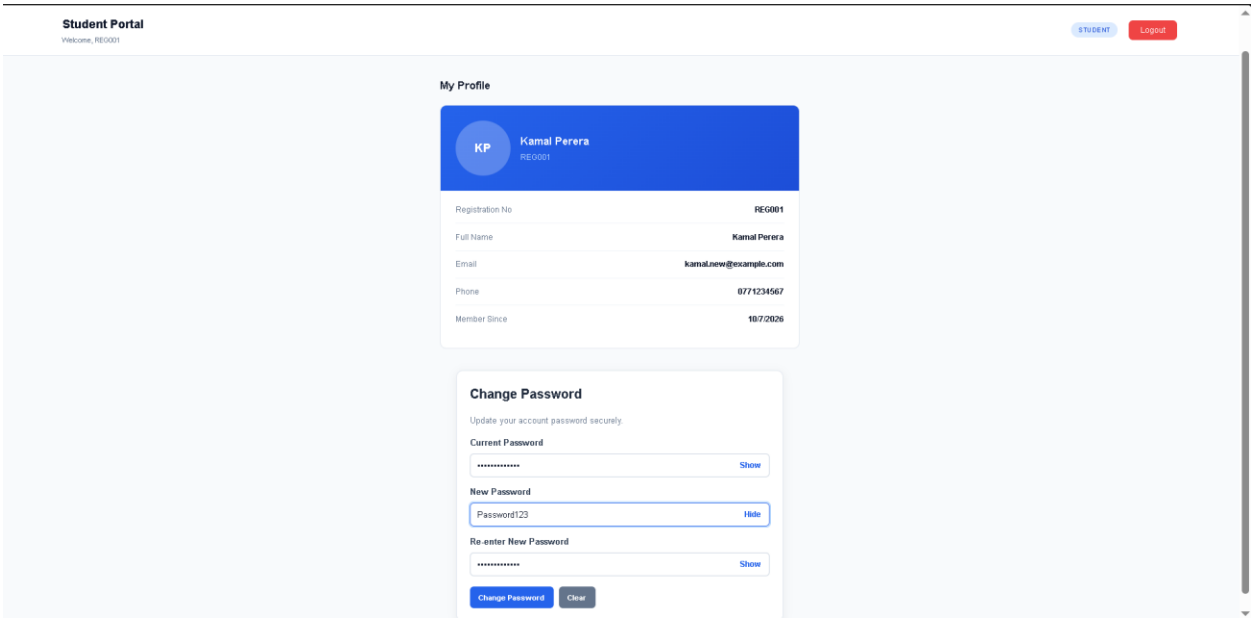


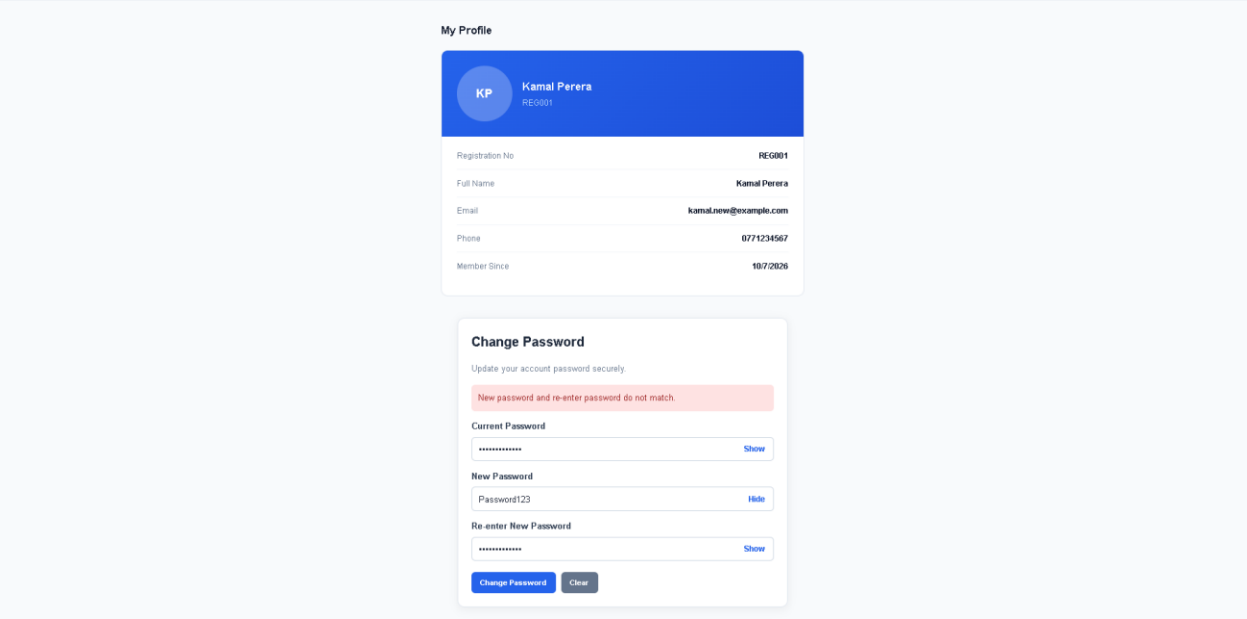
Evidence 02 – Successful Password Change and Login Verification

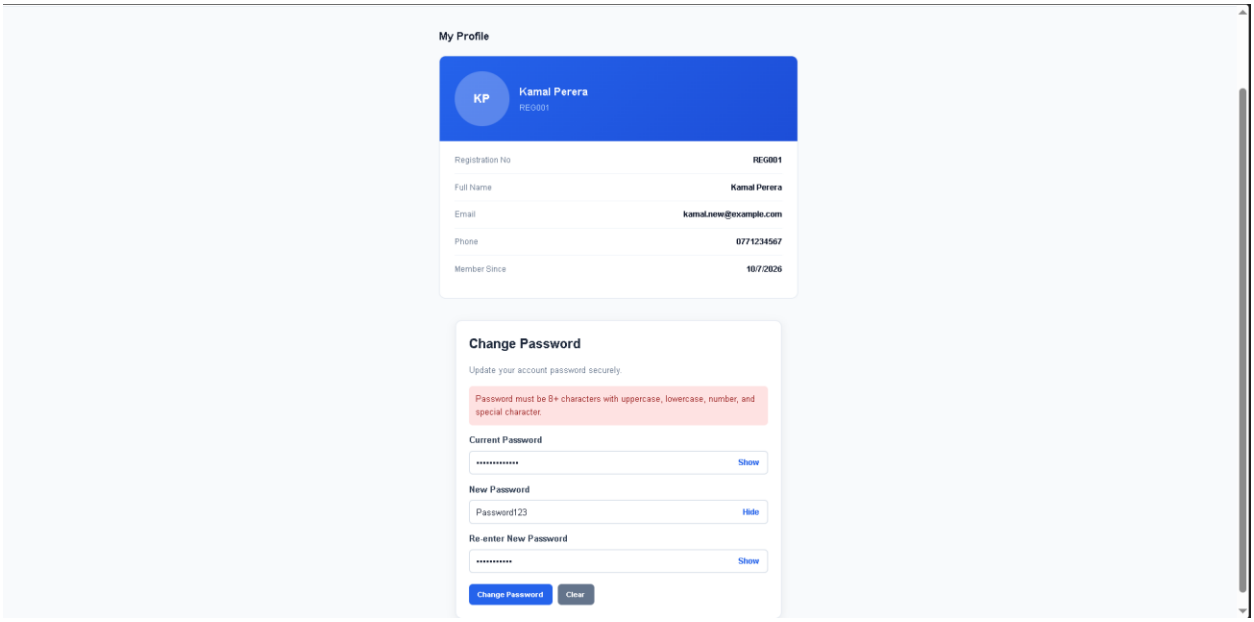
Related criteria: AC-002, AC-006, AC-008, AC-009, AC-010, AC-011

Figure 2: Password-change success notification and evidence that the new password allows login. Sensitive password values and authentication tokens must not be visible in the screenshot.


Student Portal
Welcome, REG001
STUDENT
Logout
My Profile
KP
Kamal Perera
REG001
Registration No
REG001
Full Name
Kamal Perera
Email
kamal.new@example.com
Phone
0771234567
Member Since
10/7/2026
Change Password
Update your account password securely.
Password changed successfully.
Current Password
Show
New Password
Show
Re-enter New Password
Show


Student Portal
Welcome, REG001
STUDENT
Logout
My Profile
KP
Kamal Perera
REG001
Registration No
REG001
Full Name
Kamal Perera
Email
kamal.new@example.com
Phone
0771234567
Member Since
10/7/2026
Change Password
Update your account password securely.
Current Password
Show
New Password
Password123
Hide
Re-enter New Password
Show
Change Password
Clear





6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Password Security	Verified
Existing CRUD	Verified
Final Status	Accepted

6.1 Final Summary

CR-020 was implemented to provide a secure password-change facility for authenticated students through the Student Dashboard. The implementation requires the student to enter the current password, a new password, and a confirmation password before submitting the request.

The backend verifies the current password, validates the new password against the defined strength requirements, confirms that the new password fields match, and rejects a password that is identical to the current password. The new password is hashed using bcrypt before the existing user's password hash is updated in the MySQL database.

The frontend provides appropriate success and error feedback and a Clear button to reset the password form. The password-change endpoint is protected by authentication, and the operation is restricted to the authenticated student's own account.

The implementation also preserves the existing administrator functions, student CRUD operations, and authentication structure. After successful verification of the defined test cases and acceptance criteria, CR-020 can be considered completed and accepted.

6.2 Issues / Limitations

No significant issues identified after successful implementation and verification.

Student Name: K.Varshika

Student ID: JF/ICT/24/10

Date: 13 October 2026

Signature: K.Varshika

~ END OF REPORT ~